



Edge Ai Processor

Video analytics embedded computer designed for real-time detection tasks for dynamic control of traffic light signaling and the collection of traffic data from cameras

Edge Ai Processor



Model

Dyna-NX-B000

Description

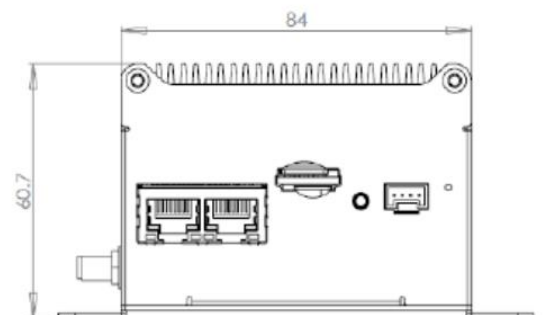
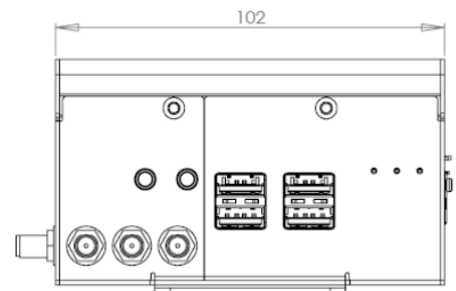
Real time video-analytics unit for installation into the traffic control cabinet

Accessories

- SDLC expander – 64 channels
- IO expander – 4/8/16 relays
- M.2 2280 M key x 1 for NVMe SSD
- M.2 3052 or 3042 B key x 1 for 5G or 4G
- M.2 2030 E key x 1 for WiFi/BT module
- V2X RSU

Other Features

- NTP time synchronization
- User management – admin, analyst, viewer
- Remote updates – over-the-air
- Data reduction profiles for remote configuration



Edge Ai Processor Specification

General Properties	
Processor	NVIDIA Jetson NX
Memory	8 GB 128-bit LPDDR4x, 16 GB eMMC 5.1
Expansion slots	M.2 2280 M Key x 1 for NVMe SSD, M.2 3052 or 3042 B key x 1 for 5G or 4G, M.2 2030 E key x 1 for WiFi/BT module
Ethernet	Supports 2 Giga LAN ports
Video output	HDMI x 1 (2.0 maximum 3840 x 2160) DP x 1
Power supply and consumption	DC 12V (3-pin terminal block), 50W
Dimensions	W84 X D102 X H54.7 mm
Gross Weight	0.3 kg
Operating / storage temperature	-20°C ~ +50°C / -40°C ~ +85°C
Storage humidity	10% ~ 90% (non-condensing)
Certification	CE/FCC Class A, according to EN 55032 & EN55035, MIL-STD-810G, Method 514.7, Category 4 MIL-STD-810G, Method 516.7, Procedure I (Shock)
Designed for installation	Traffic control cabinet / outdoor cabinet / DIN rail
Video analytics	
Video analytic engine	Exact object traces, 14 categories, in-built ALPR for LP with alphanumeric characters (EU,USA.UK.RUS), traffic light state recognition, dynamic and static anonymization, georegistration, detection masks
Processing power in FPS (B/B+A/B+LP/B+LP+A)	@544x320: 151/112/64/58; @704x419: 100/83/53/48 (B = basic detection, A = add-ons img. Processing modules, LP = license plate recognition)
Camera support	IP cameras with H.264 or H.265 codec and RTSP or ONVIF cameras / capable of processing at least 6 camera streams in the real time/ supporting narrow and wide angle cameras and cameras with motorized lens / detection range up to 120 meters
Traffic analytics	
Multifunctional engine	Fully configurable trajectory processing pipeline via visual programming language FLOW / ability to evaluate multiple detection tasks in parallel (100+)
Available filters	Zone, gate, movement, duration, time of occurrence, class, LP, color (without a limit on the number)
Other operators	Level or services, union, intersection, complement, volume
Data statistics	Incremental / whole history, time blocks, floating window, fixed interval
Outputs	Events, actions / commands, statistics, tables, histograms, images
Possible tasks	Presence detection, u-turn detection, blocking vehicle detection, red light violation, average speed, detection of specific traffic events, OD matrix, conflict detection, traffic data collection, license plate recognition
Interfaces	
Data interfaces	UDP, REST, WEBHOOK, MJPEG, XProtect (VMS-Milestone), MJPEG
HW interfaces	Support for IO expanders (relay outputs), SDLC expanders, V2X RSUS
Visual	Fully configurable dashboards with interactive widgets



DynaTech

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Dynamic Traffic System Sdn Bhd

No 52, Jalan Seroja 5D/2
Bukit Sentosa 48300 Rawang
Selangor, **MALAYSIA**

Email : sales@dynasystem.info

Authorized Agent

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